

Brilla Study Platform

Comprehensive Technical & Feature Documentation

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1. Executive Summary

Brilla Study Platform is a comprehensive educational technology solution designed specifically for Ghanaian students preparing for national examinations including NSMQ (National Science & Maths Quiz), WASSCE (West African Senior School Certificate Examination), and BECE (Basic Education Certificate Examination).

The platform combines modern web technologies with artificial intelligence to deliver personalized learning experiences, competitive practice environments, and robust progress tracking for students, teachers, and parents.

Key Highlights

- 10,000+ Practice Questions** across all major subjects
- AI-Powered Tutoring** with instant explanations and personalized guidance
- E-Library** with multimedia educational resources (PDFs, videos, audio)
- AI Student Counselor** for academic, career, and wellbeing support
- NSMQ Competition Simulation** with all 5 rounds recreated
- Live 1v1 Battles** for real-time competitive practice
- Comprehensive Analytics** for tracking student progress
- Parent Portal** for monitoring children's educational journey
- Teacher Dashboard** for class management and assessment creation

2. Platform Overview

2.1 Target Audience

User Type	Description
Students	JHS and SHS students (ages 12-18) preparing for BECE, WASSCE, and NSMQ
Teachers	Educators managing classes, creating assessments, and uploading resources
Parents	Guardians monitoring their children's academic progress
Administrators	Platform managers overseeing user management and content moderation

2.2 Supported Examinations

NSMQ (National Science & Maths Quiz)

- Full 5-round competition simulation
- Speed training for quick recall
- Riddles and problem-solving challenges
- Live 1v1 battle mode

WASSCE (West African Senior School Certificate)

- 50+ subject coverage
- Past papers from 2015-2024
- AI-powered essay grading
- Theory and objective questions

BECE (Basic Education Certificate)

- Complete JHS curriculum coverage
- All 9 core subjects
- Practice tests with progress tracking

2.3 Platform URLs

Environment	URL	Purpose
Production (Frontend)	https://brillaprep.org	Main user-facing application
Production (API)	https://brilla-api.ghwmelite.workers.dev	Backend API services
Development	localhost:5173	Local development environment

3. Technical Architecture

3.1 Technology Stack

Frontend

Technology	Version	Purpose
React	18.x	UI framework
TypeScript	5.x	Type-safe JavaScript
Vite	5.x	Build tool and dev server
Tailwind CSS	3.x	Utility-first styling
Zustand	4.x	State management
React Router	6.x	Client-side routing
Lucide React	-	Icon library

Backend

Technology	Purpose
Cloudflare Workers	Serverless compute
Hono	Web framework for Workers
Cloudflare D1	SQLite database
Cloudflare R2	Object storage for files
JWT	Authentication tokens

Technology	Purpose
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AI Integration

Provider	Model	Use Case
Anthropic	Haiku	AI Tutor, Essay Grading, Counselor

3.2 Project Structure

```
brilla-study-platform/
├── src/
│   ├── components/           # Reusable UI components
│   │   ├── auth/             # Authentication components
│   │   ├── common/           # Shared components (Button, Modal, etc.)
│   │   ├── counselor/        # AI Counselor components
│   │   ├── dashboard/        # Dashboard widgets
│   │   ├── layout/           # Layout components (Sidebar, Header)
│   │   ├── library/          # E-Library components
│   │   └── practice/         # Practice session components
│   ├── pages/                # Route page components
│   ├── stores/               # Zustand state stores
│   ├── types/                # TypeScript type definitions
│   ├── utils/                # Utility functions
│   └── data/                 # Static data and constants
├── workers/
│   └── api/                  # Cloudflare Workers API
│       ├── index.ts          # Main API router
│       ├── library.ts        # E-Library endpoints
│       └── counselor.ts      # AI Counselor endpoints
├── database/
│   └── migrations/           # D1 database migrations
├── public/                   # Static assets
└── docs/                     # Documentation
```

3.3 State Management

The platform uses Zustand for state management with the following stores:

Store	Purpose
authStore	User authentication and session management
practiceStore	Practice session state and question management
libraryStore	E-Library resources and collections
counselorStore	AI Counselor conversations
aiTutorStore	AI Tutor chat and explanations
questStore	Daily quests and achievements
battleStore	1v1 battle state
subjectStore	Subject and topic data
leaderboardStore	Rankings and leaderboard data

4. User Roles & Permissions

4.1 Role Definitions

Student

- Access practice questions and assessments
- Use AI Tutor and AI Counselor
- Browse and download E-Library resources
- Participate in competitions and battles
- Track personal progress and analytics
- Complete daily quests and earn XP

Teacher

- All student permissions
- Create and manage classes
- Build custom assessments
- Upload resources to E-Library
- View student performance reports
- Grade essay submissions

Parent

- View linked children's progress
- Access performance reports
- Monitor study activity and streaks
- Receive notifications about child's performance

Administrator

- All teacher permissions
- User management (approve, suspend, delete)
- Content moderation
- System analytics and audit logs
- Platform configuration

4.2 Permission Matrix

Feature	Student	Teacher	Parent	Admin
Practice Questions	✓	✓	-	✓
AI Tutor	✓	✓	-	✓
AI Counselor	✓	✓	-	✓
E-Library (View)	✓	✓	-	✓
E-Library (Upload)	-	✓	-	✓
E-Library (Delete Own)	-	✓	-	✓
E-Library (Delete Any)	-	-	-	✓
Create Assessments	-	✓	-	✓
View Child Progress	-	-	✓	-
User Management	-	-	-	✓
Audit Logs	-	-	-	✓

5. Core Features

5.1 Authentication System

Registration Flow

1. User selects role (Student, Teacher, Parent)

- 2. Provides email, password, and profile information
- 3. Students/Teachers select school and grade level
- 4. Account created with email verification (optional)
- 5. Teachers/Students require admin approval

Login Methods

- Email and password
- Persistent sessions with JWT tokens
- Automatic session refresh

Security Features

- Password hashing with bcrypt
- JWT token expiration (7 days)
- Role-based access control
- Account suspension capability

5.2 Dashboard

Each user role has a customized dashboard:

Student Dashboard

- XP and level progress
- Current streak status
- Daily quest progress
- Quick access to practice
- Recent activity feed
- Upcoming assessments

Teacher Dashboard

- Class overview
- Pending assignments to grade
- Student performance summary
- Quick actions for common tasks

Parent Dashboard

- Children's progress overview
- Recent activity timeline
- Performance trends
- Study time statistics

5.3 Subject & Topic System

The platform organizes content into a hierarchical structure:

```
Subject (e.g., Mathematics)
├── Topic (e.g., Algebra)
│   └── Questions (Multiple types)
```

Supported Subjects

- Mathematics
- English Language
- Integrated Science
- Social Studies
- French
- ICT
- Physics
- Chemistry
- Biology
- Geography
- History
- Economics
- And 40+ more subjects

5.4 Question Types

Type	Description	Supported Exams
Multiple Choice	4 options, single correct answer	All

True/False Type	Binary choice questions Description	All Supported Exams
Fill in the Blank	Text input for missing words	All
Short Answer	Brief text responses	WASSCE, BECE
Essay	Long-form written responses with AI grading	WASSCE
Speed Round	Time-limited questions (NSMQ style)	NSMQ
Riddle	Word puzzles and brain teasers	NSMQ

6. E-Library System

6.1 Overview

The E-Library is a comprehensive digital resource management system allowing teachers and administrators to upload, organize, and share educational materials with students.

6.2 Supported Resource Types

Type	Extensions	Max Size	Features
PDF	.pdf	50 MB	In-browser viewing, progress tracking
Video	.mp4, .webm, .mov	500 MB	Streaming playback, progress tracking
Audio	.mp3, .wav, .m4a	100 MB	Audio player with controls
Document	.doc, .docx, .txt	20 MB	Download only
Interactive	.html, .zip	50 MB	Embedded iframe
Link	URL	-	External resource linking

6.3 Resource Management

Upload Process

1. Select resource type
2. Drag and drop or browse for file
3. Enter metadata (title, description, subject, tags)
4. Set access level (Free, Basic, Premium)
5. Configure downloadable option
6. Optional: Mark as featured
7. Submit for processing

Resource Metadata

- Title (required)
- Description
- Resource type
- Subject association
- Topic association
- School level (JHS, SHS, Both)
- Access level
- Tags (comma-separated)
- Featured flag
- Downloadable flag

6.4 Collections

Users can organize resources into personal collections:

- Create named collections
- Add resources to multiple collections
- Reorder resources within collections

- Share public collections

6.5 Features

Feature	Description
Search	Full-text search across titles, descriptions, and tags
Filtering	Filter by type, subject, school level, access level
Sorting	Sort by newest, popular, rating, or title
Progress Tracking	Track viewing/reading progress per resource
Ratings & Reviews	5-star rating system with optional reviews
View/Download Counts	Analytics on resource popularity
Bookmarking	Save resources for quick access

6.6 API Endpoints

Method	Endpoint	Description
GET	/api/library/resources	List/search resources
GET	/api/library/resources/:id	Get resource details
GET	/api/library/featured	Get featured resources
POST	/api/library/upload	Upload new resource
PUT	/api/library/resources/:id	Update resource
DELETE	/api/library/resources/:id	Delete resource
GET	/api/library/files/*	Serve file from R2

7. AI Student Counselor

7.1 Overview

The AI Student Counselor provides personalized support to students across three specialized domains: academic guidance, career counseling, and wellbeing support.

7.2 Counselor Types

Academic Counselor

- Study strategy recommendations
- Subject selection guidance
- Exam preparation tips
- Time management advice
- Learning difficulty support

Career Counselor

- STEM career exploration
- University program guidance
- Scholarship information
- Career path planning
- Skills development advice

Wellbeing Counselor

- Stress management techniques
- Study-life balance guidance
- Motivation and confidence building
- Anxiety and pressure coping
- Mental health resources (referral when appropriate)

7.3 Features

Feature	Description
Conversation History	Persistent chat history across sessions
Context Awareness	Uses student's performance data for personalized advice
Resource Suggestions	Recommends relevant E-Library materials
Typing Animation	Natural message delivery experience
Feedback System	Rate message helpfulness
Quick Actions	Pre-defined prompts for common questions

7.4 Safety Measures

- Sentiment detection for concerning messages
- Escalation protocols for crisis situations
- Professional referral recommendations
- Conversation flagging for admin review

7.5 API Endpoints

Method	Endpoint	Description
GET	/api/counselor/conversations	List user's conversations
POST	/api/counselor/conversations	Start new conversation
GET	/api/counselor/conversations/:id	Get conversation with messages
POST	/api/counselor/chat	Send message and get AI response
DELETE	/api/counselor/conversations/:id	Archive conversation

8. Practice & Assessment System

8.1 Practice Modes

Topic Practice

- Select subject and topic
- Unlimited practice questions
- Immediate feedback with explanations
- AI Tutor available for help

Timed Practice

- Configurable time limits
- Question count selection
- Performance scoring
- Speed metrics tracking

Mock Exams

- Full exam simulation
- Realistic timing
- Question paper format
- Comprehensive results

Past Papers

- WASSCE papers (2015-2024)
- BECE papers
- Original marking schemes
- AI essay grading

8.2 Assessment Builder

Teachers can create custom assessments with:

- Multiple question types
- Time limits
- Randomized question order
- Scheduled availability
- Class/group assignment
- Grading rubrics

8.3 AI-Powered Essay Grading

For essay-type questions:

1. Student submits essay response
2. AI analyzes content, structure, grammar
3. Provides score based on rubric
4. Generates detailed feedback
5. Suggests improvements

8.4 Progress Tracking

Metric	Description
Accuracy	Percentage of correct answers
Speed	Average time per question
Streak	Consecutive days of practice
XP	Experience points earned
Level	Current proficiency level
Topics Mastered	Topics with >80% accuracy

9. Competition & Gamification

9.1 NSMQ Competition Mode

Full simulation of the National Science & Maths Quiz:

Round 1: Fundamental

- Basic recall questions
- 30 seconds per question
- Individual scoring

Round 2: Speed Race

- First to answer correctly wins
- 10 seconds time limit
- Bonus points for speed

Round 3: Problem of the Day

- Complex multi-step problem
- Extended time allowed
- Partial credit available

Round 4: True or False

- Rapid-fire format
- 15 seconds per question
- Penalty for wrong answers

Round 5: Riddles

- Word puzzles and brain teasers
- Lateral thinking required
- Point multipliers

9.2 Live 1v1 Battles

Real-time competitive mode:

- Matchmaking by skill level
- Synchronized questions
- Live score updates
- Victory/defeat animations
- XP rewards

9.3 House System

School-based team competition:

- Join your school house
- Contribute to house points
- Weekly house rankings
- House Cup championship

9.4 Leaderboards

Leaderboard	Scope	Ranking Criteria
National	All users	Total XP
School	Same school	Weekly XP
Subject	Per subject	Accuracy + XP
Battle	Battle participants	Win rate

9.5 Gamification Elements

XP System

Action	XP Reward
Answer correctly	10 XP
Complete topic	50 XP
Daily streak	25 XP
Win battle	100 XP
Complete quest	Variable

Achievements

- First Steps: Complete first practice
- Streak Master: 30-day streak
- Battle Champion: Win 50 battles
- Subject Expert: Master all topics in a subject
- Speed Demon: Answer 100 questions under 5 seconds

Daily Quests

- Practice X questions
 - Complete X topics
 - Achieve X% accuracy
 - Win X battles
 - Use AI Tutor
-

10. Analytics & Reporting

10.1 Student Analytics

Performance Dashboard

- Overall accuracy trend graph
- Subject-by-subject breakdown
- Strength and weakness analysis
- Time spent studying
- Comparison to peers (percentile)

Progress Reports

- Weekly summary emails
- Monthly detailed reports
- Exportable PDF reports

10.2 Teacher Analytics

Class Performance

- Class average scores
- Individual student progress
- Topic coverage heatmap
- Assessment results analysis

Resource Engagement

- View counts per resource
- Download statistics
- Student completion rates

10.3 Admin Analytics

Platform Metrics

- Daily/weekly/monthly active users
- Question attempt volume
- Popular subjects and topics
- User growth trends

Audit Logs

- User actions tracking
 - Content modifications
 - System changes
 - Security events
-

11. Parent Portal

11.1 Features

Child Linking

- Request link to child's account
- Child approves connection
- Multiple children support

Progress Monitoring

- Real-time activity feed
- Practice session summaries
- Assessment results
- Streak status

Reports

- Weekly email summaries
- Performance trend charts
- Study time analysis
- Subject-wise breakdown

Notifications

- Low activity alerts
- Achievement celebrations
- Assessment reminders
- Streak at-risk warnings

11.2 Settings

- Notification preferences
 - Report frequency
 - Privacy controls
 - Child account management
-

12. Teacher Dashboard

12.1 Class Management

Create Classes

- Class name and description
- Grade level selection
- Subject assignment
- Enrollment settings

Student Management

- View enrolled students
- Add/remove students
- Assign groups within class
- View individual performance

12.2 Assessment Tools

Assessment Builder

- Question bank access
- Custom question creation
- Multiple assessment types
- Scheduling options

Grading

- Auto-graded objectives
- Manual essay grading
- AI-assisted essay scoring
- Feedback templates

12.3 Content Upload

E-Library Contributions

- Upload educational resources
 - Organize by subject/topic
 - Set access permissions
 - Track engagement
-

13. Admin Panel

13.1 User Management

User Operations

- View all users
- Search and filter
- Edit user details
- Change user roles
- Suspend/activate accounts
- Delete accounts

Approval Workflow

- Pending teacher approvals
- Pending student approvals
- Bulk approval actions

13.2 Content Management

Question Management

- Add/edit/delete questions
- Import questions (CSV)
- Question quality review
- Tag management

Resource Moderation

- Review flagged content
- Approve/reject uploads
- Featured content curation

13.3 System Settings

Platform Configuration

- Feature toggles
- Rate limits
- Notification settings
- Maintenance mode

Analytics Access

- Full platform analytics
- Export capabilities
- Custom report generation

13.4 Audit System

- Complete action logging
- User activity trails
- Security event monitoring
- Data export for compliance

14. Database Schema

14.1 Core Tables

users

```
CREATE TABLE users (
  id TEXT PRIMARY KEY,
  email TEXT UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,
  name TEXT NOT NULL,
  role TEXT CHECK (role IN ('student', 'teacher', 'admin')),
  school TEXT,
  grade_level TEXT,
  house TEXT,
  xp INTEGER DEFAULT 0,
  level INTEGER DEFAULT 1,
  streak_days INTEGER DEFAULT 0,
  longest_streak INTEGER DEFAULT 0,
  last_practice_date TEXT,
  is_verified INTEGER DEFAULT 0,
  is_approved INTEGER DEFAULT 0,
  created_at TEXT DEFAULT (datetime('now'))
);
```

library_resources

```
CREATE TABLE library_resources (
  id TEXT PRIMARY KEY,
  title TEXT NOT NULL,
  description TEXT,
  resource_type TEXT NOT NULL,
  content_url TEXT NOT NULL,
  thumbnail_url TEXT,
  file_size INTEGER,
  duration INTEGER,
  subject_id TEXT,
  topic_id TEXT,
  school_level TEXT,
  access_level TEXT DEFAULT 'free',
  tags TEXT,
  uploaded_by TEXT NOT NULL,
  is_featured INTEGER DEFAULT 0,
  is_active INTEGER DEFAULT 1,
  is_downloadable INTEGER DEFAULT 1,
  views INTEGER DEFAULT 0,
  downloads INTEGER DEFAULT 0,
  created_at TEXT DEFAULT (datetime('now'))
);
```

counselor_conversations

```
CREATE TABLE counselor_conversations (
  id TEXT PRIMARY KEY,
  user_id TEXT NOT NULL,
  counselor_type TEXT NOT NULL,
  title TEXT,
  status TEXT DEFAULT 'active',
  message_count INTEGER DEFAULT 0,
  last_message_at TEXT,
  created_at TEXT DEFAULT (datetime('now'))
);
```

14.2 Migration History

Migration	Description
	User verification

001 Migration	system Description
002	Parent system
003	Audit logging
004	Assessment system
005	Quests system
006	Streak protection
007	Reminders system
008	E-Library system
009	AI Counselor
010	Counselor reports
011	Downloadable field

15. API Reference

15.1 Authentication

Register

```
POST /api/auth/register
Content-Type: application/json

{
  "email": "student@example.com",
  "password": "securepassword",
  "name": "John Doe",
  "role": "student",
  "school": "Presec Legon",
  "gradeLevel": "SHS 2"
}
```

Login

```
POST /api/auth/login
Content-Type: application/json

{
  "email": "student@example.com",
  "password": "securepassword"
}

Response:
{
  "success": true,
  "data": {
    "user": { ... },
    "token": "jwt-token-here"
  }
}
```

15.2 Protected Routes

All protected routes require:

Authorization: Bearer <jwt-token>

15.3 Response Format

Success:

```
{
  "success": true,
  "data": { ... },
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 100,
    "totalPages": 5
  }
}
```

Error:

```
{
  "success": false,
  "error": "Error message here"
}
```

16. Deployment & Infrastructure

16.1 Cloudflare Services

Service	Resource	Purpose
Pages	brilla-study-platform	Frontend hosting
Workers	brilla-api	Backend API
D1	brilla-db	SQLite database
R2	brilla-library	File storage

16.2 Environment Variables

Frontend (Cloudflare Pages)

Variable	Description
VITE_API_URL	Backend API URL

Backend (Cloudflare Workers)

Variable	Description
ENVIRONMENT	production/development
JWT_SECRET	JWT signing secret
ANTHROPIC_API_KEY	AI API key
AI_PROVIDER	AI service provider

AI_MODEL Variable	AI model identifier Description
----------------------	------------------------------------

16.3 Deployment Process

Frontend

```
npm run build
npx wrangler pages deploy dist --project-name=brilla-study-platform
```

Backend

```
npx wrangler deploy
```

Database Migrations

```
npx wrangler d1 execute brilla-db --remote --file=database/migrations/XXX.sql
```

17. Security Considerations

17.1 Authentication Security

- Passwords hashed with bcrypt (salt rounds: 10)
- JWT tokens with 7-day expiration
- Secure HTTP-only cookies option
- Rate limiting on auth endpoints

17.2 Data Protection

- HTTPS enforced in production
- Input validation on all endpoints
- SQL injection prevention via parameterized queries
- XSS protection through React's built-in escaping

17.3 Access Control

- Role-based permission system
- Resource ownership verification
- Admin approval for sensitive roles
- Audit logging for accountability

17.4 File Security

- File type validation on upload
- Size limits per resource type
- Virus scanning (recommended)
- Secure signed URLs for downloads

18. Future Roadmap

Phase 1: Enhanced AI Features

- Voice-based AI tutoring
- Handwriting recognition for math
- Personalized study plan generation
- Predictive performance analytics

Phase 2: Social Features

- Study groups
- Peer tutoring
- Discussion forums
- Collaborative note-taking

Phase 3: Mobile Applications

- Native iOS application
- Native Android application
- Offline mode support
- Push notifications

Phase 4: Institutional Features

- School dashboard
- District-level analytics
- Curriculum alignment tools
- LMS integration (Canvas, Moodle)

Phase 5: Content Expansion

- More past papers
- Video explanations for all topics
- Interactive simulations
- Augmented reality labs

Appendix A: Glossary

Term	Definition
XP	Experience Points earned through platform activities
Streak	Consecutive days of practice
House	School-based team for competition
Quest	Daily or weekly challenge with rewards
Battle	Real-time 1v1 competitive practice session
Mock Exam	Full simulation of actual examination
AI Tutor	Intelligent assistant for explanations
AI Counselor	Virtual advisor for academic/career/wellbeing

Appendix B: Contact & Support

Technical Support: For platform issues, contact the development team at Hodges & Co.

Content Inquiries: For questions about educational content, contact platform administrators.

Security Reports: For security vulnerabilities, report through responsible disclosure channels.

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Document End